Active Reconnaissance and Vulnerability Scanning Report

1. Objective
The objective of this practical exercise is to identify open ports, running services, and potential security vulnerabilities on an authorized test system using Nmap, Nikto, and OpenVAS.

2. Tools Used
- Nmap: Port scanning and service enumeration.
- Nikto: Web server security checks.
- OpenVAS: Vulnerability assessment.

3. Methodology
First, Nmap was used to scan the authorized target and identify open ports and running services. Next, Nikto was used to check the local test web server for potentially insecure configurations and known risky files. Finally, OpenVAS was used to assess the target for known vulnerabilities.

4. Findings
The scan findings must be completed using the actual results obtained during testing.

- Open ports: Record the ports identified by Nmap.
- Services detected: Record the service names and versions.
- Vulnerabilities: Record the findings reported by Nikto and OpenVAS.
- Severity: Classify confirmed findings as Critical, High, Medium, Low, or Informational according to the scanner results.

5. Security Risks
Unnecessary open ports may expose services to unauthorized access. Outdated software may contain known vulnerabilities. Insecure web server configurations can increase the risk of information disclosure or unauthorized activity. Actual risk depends on the system configuration and whether the findings can be exploited.

6. Recommendations
Close unnecessary ports, disable unused services, update vulnerable software, apply security patches, improve web server configuration, and repeat the scan after remediation.

7. Conclusion
Active reconnaissance and vulnerability scanning help identify potential security weaknesses before attackers can exploit them. Nmap, Nikto, and OpenVAS provide complementary information that supports security assessment and remediation.

Note: Replace the findings section with the actual scan results before submitting the report.